QEA-FAD: Quantum Enhanced Autoencoders For Anomaly Detection in Finance and Healthcare

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2025 International Conference on Networks and Cryptology (NETCRYPT) | 979-8-3315-2605-4/25/$31.00 ©2025 IEEE | DOI: 10.1109/NETCRYPT65877.2025.11102801

***Abstract*—Quantum Auto-Encoders (QAEs) represent a transformative approach to anomaly detection, leveraging quantum computational power to address challenges in domains such as healthcare and finance. In this study, we focus on detecting anomalies in breast cancer diagnosis and fraudulent financial transactions. QAEs are trained exclusively on non-anomalous data, enabling anomaly detection through their inability to faithfully reconstruct anomalous points. Reconstruction quality is evaluated using fidelity tests for simulations and SWAP tests for quantum hardware, ensuring reliable and robust performance across diverse environments. Our work introduces QEA-FAD: Quantum Enhanced Autoencoders for Anomaly Detection, incorporating enhanced and patch-based architectures that demonstrate significant advancements in precision. By integrating quantum entanglement into the encoding and decoding processes, via shared Bell pairs and direct qubit entanglement. Our framework improves correlations and delivers statistically significant performance gains.**

Benchmarks on the MNIST dataset validate the scalability of our approach, while JAX-accelerated optimization reduces training time on financial datasets. Experiments on IBM and Amazon quantum hardware confirm the practical feasibility of QAEs. These advancements highlight the efficiency of QAEs, achieving superior accuracy with reduced computational resources, making them a compelling solution for real-world anomaly detection challenges in critical applications.

***Index Terms*—Quantum Autoencoders (QAEs), Quantum Anomaly Detection, Quantum Entanglement, Reconstruction, Fidelity, Quantum Machine Learning (QML)**

I. Introduction

Anomaly detection is a critical task in machine learning (ML), with applications spanning network monitoring, medical diagnosis, and financial fraud detection [1]–[5]. The inherent challenges in this field include the scarcity of labeled anomalous data and the imbalance between normal and anomalous instances [1], [6], [7]. Traditional supervised methods, while effective in some scenarios, often fail when anomalies occur infrequently or remain unidentified during training. Unsupervised and semi-supervised approaches, which rely on training models exclusively on normal data, have emerged as viable alternatives [2]. Among these, reconstruction-based models like autoencoders are widely utilized, detecting anomalies by identifying substantial deviations in reconstructed data compared to the input [7]–[10].

Quantum Machine Learning (QML), which combines the principles of quantum computing and ML, offers a transformative approach to anomaly detection by leveraging the unique properties of quantum systems such as superposition and entanglement [8]–[15]. Quantum Autoencoders (QAEs) extend the classical autoencoder framework into the quantum domain, encoding data into compressed quantum states and decoding them for reconstruction [16], [17]. Anomalies are detected either through analysis of the latent quantum states or by evaluating reconstruction errors. These methods are particularly suited for scenarios where labeled anomalous data is unavailable, enabling models to generalize effectively from normal training datasets.

This paper investigates QAEs for anomaly detection across diverse datasets, utilizing fidelity tests in simulations and SWAP tests on quantum hardware for robust evaluation [16], [17]. By benchmarking against classical autoencoders [3], we demonstrate the scalability and efficiency of QAEs in tackling imbalanced and complex data distributions. Experimental results highlight the potential of quantum computing to address critical ML challenges, paving the way for broader adoption of quantum techniques in real-world anomaly detection tasks.

A. Contributions

The following concisely describes the primary contributions of this work:

- This paper introduces *QEA-FAD*, a novel framework that enhances quantum autoencoders by incorporating quantum entanglement to optimize the correlation between encoding and decoding processes. The model leverages entangled encoder-decoder qubits and shared Bell states to improve reconstruction fidelity, achieving higher accuracy in anomaly detection tasks.
- A hybrid evaluation methodology is proposed, combining fidelity tests for classical simulations and SWAP tests for execution on quantum processing units (QPUs). This dual approach ensures robust assessment of QEA-FAD across both simulated environments and real quantum hardware,

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emphasizing its adaptability to noisy intermediate-scale quantum (NISQ) devices.

- Extensive benchmarking against classical autoencoders demonstrates the superiority of QEA-FAD, which achieves comparable or better anomaly detection performance while requiring fewer parameters and training iterations. This highlights the computational efficiency and scalability of the model in addressing high-dimensional and imbalanced datasets.
- The effectiveness of QEA-FAD is validated through real-world applications, including breast cancer diagnosis in healthcare and fraud detection in financial transactions. The results showcase the model's ability to deliver high detection accuracy, reinforcing its potential as a practical tool for solving critical anomaly detection problems.

B. Organisations

The remaining segments of the manuscript are arranged as follows: Section II provides a review of relevant background pertaining to our research. In Section III, we outline the proposed methodology. Section IV presents the observed results and findings. Finally, Section V offers concluding remarks and the future scope.

II. Background studies

The use of quantum computing for anomaly detection has garnered significant attention, with quantum autoencoders (QAEs) emerging as a promising approach. Early foundational work by authors in [16], [17] introduced fully quantum autoencoders, paving the way for subsequent advancements. For instance, researchers in [18] proposed hybrid classical-quantum autoencoders, integrating quantum variational circuits into the latent space, while [19] replaced classical layers in deep learning models with quantum variational circuits. In high-energy physics, scientists in [20] employed QAEs for anomaly detection using both simulators and real quantum hardware, albeit with limited features. Similarly, authors in paper [21] demonstrated anomaly detection using QAEs on synthetically generated datasets. Beyond anomaly detection, QAEs have also been applied to data compression in cloud processing, as highlighted by researchers [22]. Despite these advances, the application of QAEs to established and complex benchmarks in anomaly detection, particularly in healthcare and finance, remains limited. This work bridges the gap by leveraging QEA-FAD to address real-world anomaly detection challenges, demonstrating scalability and effectiveness across critical domains.

III. Proposed Methodologies

In this section, we present a novel anomaly detection framework utilizing Quantum Autoencoders (QAEs). This methodology is structured around four key components: (i) Training QAEs exclusively on non-anomalous data to identify deviations through suboptimal reconstructions, (ii) Evaluating reconstruction fidelity via simulations and SWAP tests on quantum hardware, (iii) Enhancing detection accuracy with patch-based and entanglement-assisted autoencoder architectures to strengthen encoding-decoding correlations, and (iv) Benchmarking the models on healthcare and finance datasets to demonstrate practical applicability. To provide a comprehensive understanding, we integrate computational optimizations through multi-processing techniques and validate experiments on real quantum platforms.

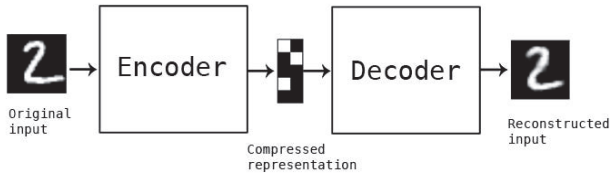

Fig. 1. An ilustration of an autoencoder taking the MNIST dataset as input.

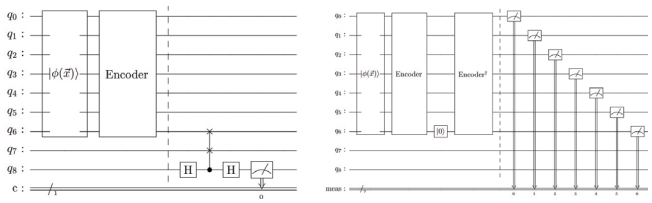

Fig. 2. a) Trainable QAE Design; b) Optimized QAE Architecture

A. Quantum Autoencoders (QAEs)

An autoencoder is a neural network designed to discover structures within data and reduce its dimensionality. It consists of two parts: the encoder, which transforms the data into a new representation, and the decoder, which reconstructs the original input from this representation, with the aim of minimizing the difference between the reconstructed and original input. In classical autoencoders, both the encoder and decoder are usually fully-connected neural networks (see Fig. 1).

QAEs are a specialized quantum machine learning model designed for tasks like data compression, representation learning, and anomaly detection. They draw inspiration from classical autoencoders, adapting their architecture to quantum computing frameworks. The QAE workflow comprises several critical stages: state preparation, encoding, qubit reset, decoding, and training, each contributing to the efficient handling of quantum data.

1) State Preparation: The anomaly detection process begins with transforming input data into a valid quantum state using amplitude encoding. This approach efficiently encodes a large number of input features using only log(N) qubits for N data points, significantly reducing resource requirements. Each data window w_i is mapped to a quantum state $|\phi_i\rangle$, where the components of w_i determine the probability amplitudes of the quantum state. This deterministic, parameter-free preparation step ensures that the input data aligns with quantum processing requirements, allowing a seamless transition into the encoding phase.

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2) Encoding and Compression: The quantum encoder applies variational quantum gates to project the high-dimensional quantum state into a compressed latent representation. For instance, in a 7-qubit architecture, the encoder maps the 7-dimensional Hilbert space into a 6-dimensional space by condensing information into qubits q_0 to q_5. Qubit q_6, designated as the "trash qubit," is reset to the zero state, ensuring that it no longer carries any meaningful information. The compression achieved in this step is pivotal for detecting deviations in the reconstruction process, which signifies anomalies. (see Fig. 2)

3) Decoding and Reconstruction: The decoding process leverages the unitary nature of the encoding transformation. The decoder, defined as the conjugate transpose of the encoder, reconstructs the input state by reversing the compression process. Ideally, the reconstructed quantum state matches the original, barring any loss of information during encoding. Discrepancies between the original and reconstructed states, quantified through fidelity tests or SWAP tests, reveal potential anomalies. This unitary inversion minimizes the need for additional training, contrasting classical approaches.

4) Training and Evaluation: The training process optimizes the encoder to map relevant data features into the latent qubits while minimizing information loss. A swap-test circuit, incorporating ancillary qubits, evaluates the reconstruction fidelity by comparing the original and reconstructed states. The probability of obtaining specific outcomes (e.g., $|0\rangle$ on the ancilla qubit) provides a measure of fidelity. Training continues until the encoder reliably resets the trash qubits and achieves maximum fidelity for normal data. Anomalous data, diverging significantly from this learned distribution, results in lower fidelity scores, effectively signaling anomalies.

B. QEA-FAD workflow

The QEA-FAD draws inspiration from classical autoencoders but leverages variational quantum circuits for its encoder and decoder instead of fully-connected neural networks. The input to QEA-FAD is a quantum state, represented here as the $|0000\rangle$ state. Classical data is embedded into the quantum circuit using techniques such as angle embedding and amplitude embedding, both of which ensure efficient mapping of high-dimensional data into quantum states. In this approach, E represents the quantum encoder, while $D = E^\dagger$ (the conjugate transpose of E) serves as the decoder. During encoding, some qubits are measured and discarded as "trash qubits," while the remaining "latent space qubits" are passed to the decoder. The decoder takes these latent qubits along with newly initialized qubits in the state $|00...00\rangle$ as inputs. (see Fig. 3)

The training process optimizes both the encoder and decoder variational circuits so that the measured trash qubits align with the initialized qubits in the decoder. Ideally, if the trash qubits are in the state $|00...00\rangle$, the output qubits of the decoder should perfectly reconstruct the original input state $|0000\rangle$. This compression process mirrors the functionality of a classical autoencoder, where the latent qubits represent a compact encoding of the input quantum state. By preserving essential data features in the latent space, QEA-FAD achieves efficient anomaly detection with robust quantum representation.

Fig. 3. An example of a quantum autoencoder.

C. Anomaly Detection with Autoencoder

Autoencoders are commonly used for anomaly detection by training on normal, non-anomalous data, enabling the model to reproduce this data with a high degree of similarity. During inference, if the autoencoder's reconstruction of a sample data point closely matches the similarity achieved during training, the data is classified as non-anomalous. Conversely, if the reconstruction deviates significantly, the data is considered anomalous, likely originating from a different distribution.

In a quantum autoencoder, anomaly detection involves comparing the state of the "trash qubits" to the $|00...00\rangle$ state. Similarity (or fidelity) between these quantum states can be measured using techniques such as the swap test on quantum hardware or direct fidelity calculation in classical simulations.

For example, in the context of medical diagnostics, a quantum autoencoder could be trained on symptom data from healthy patients. When a new patient's symptoms are input, the fidelity between the trash qubits' state and the $|00...00\rangle$ state is calculated. A low fidelity suggests the patient may be infected, indicating an anomaly, while high fidelity suggests the patient is healthy.

D. Datasets for Model Assessment

1) Breast Cancer Diagnosis:- Decoding Cellular Features: The Breast Cancer Wisconsin (Diagnostic) Dataset, contains 569 samples classified as malignant or benign. Each sample is described by 30 numerical features derived from digitized images of fine needle aspirates (FNA) of breast masses. To enhance the dataset for quantum processing, we engineered two additional features; 'under average' and 'over average', representing deviations beyond two standard deviations. This augmentation, increasing the total features to 32, strongly correlated with disease classification. Experiments used all 32 features or the top 8 most relevant ones for efficiency.

2) Fraud Detection:- Unmasking Anomalies in Transactions: The Credit Card Fraud Detection Dataset, sourced from Kaggle, comprises 284,807 transactions with 492 labeled as fraudulent (Class 1) and the rest non-fraudulent (Class 0). Its 30 features, derived via principal component analysis (PCA), ensure privacy while retaining transaction patterns. Additionally, the dataset includes 'Time,' indicating elapsed seconds between transactions, and 'Amount,' representing transaction value. Its imbalanced nature provides a challenging testbed for

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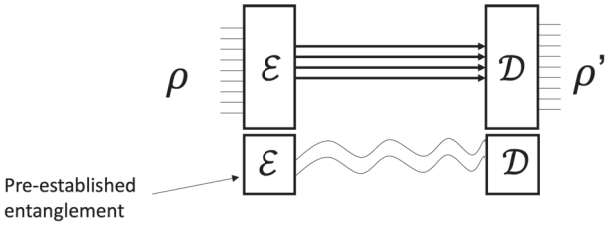

Fig. 4. Pre-encoding entanglement setup in quantum autoencoding.

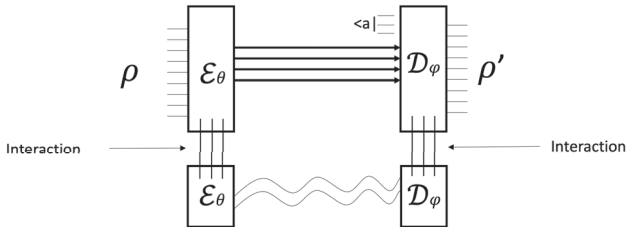

Fig. 5. QEA-FAD Model Featuring Entangled Encoder Qubits.

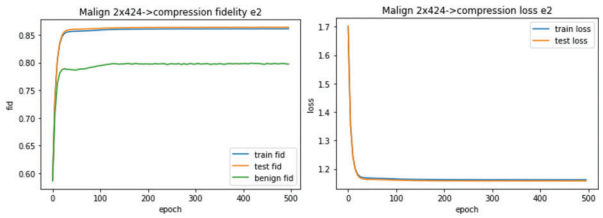

Fig. 6. a) Compression Fidelity & b) Compression Loss; Over Epochs.

evaluating the Quantum Enhanced Autoencoder in detecting rare fraudulent activities.

E. Design & Fabrication of QEA-FAD: A Novel Approach

1) Entanglement-Driven Quantum Autoencoder: Quantum entanglement serves as a powerful resource that enhances computational and communicative tasks beyond the capabilities of classical systems. This advantage is evident in scenarios such as the CHSH game, where entanglement improves winning probabilities, and quantum superdense coding, where it boosts communication rates. In this work, we explore the integration of entanglement into quantum autoencoding tasks.

QEA-FAD compresses quantum information into lower-dimensional Hilbert spaces, and prior research suggests that entanglement resources can enhance this compression process. Building on state-of-the-art QAE techniques, we incorporate entanglement resources to investigate potential improvements in compression efficiency. The framework involves establishing entanglement prior to the encoding phase, enabling coordination between the sender and receiver during encoding and decoding. The specific utilization of entanglement varies depending on the implementation, leading to diverse outcomes. Future efforts will focus on testing this approach with datasets that challenge traditional quantum autoencoder schemes, where standard methods fail to achieve effective compression and decompression. We hypothesise that incorporating entanglement in such cases will yield significant performance gains.

Fig. 4 depicts the role of entanglement within the quantum autoencoding process, highlighting its pre-encoding setup & functional integration into encoding and decoding. In our work, we tested two approaches: One in which the encoder interacts with with entangled qubits (see Fig. 5) and the other in which the encoder does not interact with the entangled qubits.

The first approach involved interacting data qubits with entangled qubits by extending the encoding ansatz to include entangled bits, using the Bell state as the coding qubits. However, this approach was too restrictive and performed worse than without entanglement. A second attempt to avoid qubit interaction had minimal effect. Further research revealed that entanglement can serve as an ideal coordinator or, with additional classical resources, enable quantum teleportation, reducing quantum transmission requirements. Without classical information, pre-measurement outcomes guide the encoding and decoding process. This is achieved through mid-circuit measurements and classical control gates on the encoder and decoder.

IV. Experimental Results and Analysis

A. Performance Evaluation in Breast Cancer Detection

In this study, we evaluated the performance of three distinct quantum circuit ansatz for implementing a quantum autoencoder aimed at detecting anomalies in the Breast Cancer Dataset. The first ansatz represents our novel proposed design, while the second and third ansatz are derived from established approaches in prior related works, serving as comparative benchmarks. This comparative analysis highlights the effectiveness and potential improvements offered by our proposed ansatz in detecting anomalies within the dataset.

1) First ansatz: Our approach draws inspiration from the Scalable Variational Quantum Encoder framework, leveraging the concept of dual quantum autoencoders to process data. Specifically, we adapted the architecture introduced in the foundational quantum autoencoder work, which we refer to as e2. However, our method incorporates novel modifications in the entire autoencoding process.

For the first encoder, we employed amplitude embedding to encode the data, while the second encoder utilized angle embedding. This dual-embedding strategy introduces complementary perspectives in data representation, enhancing the model's performance.

To further improve results, we designed an interaction mechanism between the two encoders by introducing two CNOT gates. These gates connect the latent space of one encoder to the trash qubit of the other (see Fig. 5), facilitating enhanced coordination between the encoders. The results (see Fig. 6) of these experiments, presented below, demonstrate the effectiveness of this modified circuit in achieving superior outcomes.

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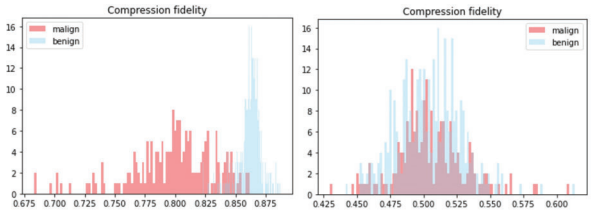

Fig. 7. a) Qiskit Simulation & b) IBM QPU Hardware; results of QEA-FAD.

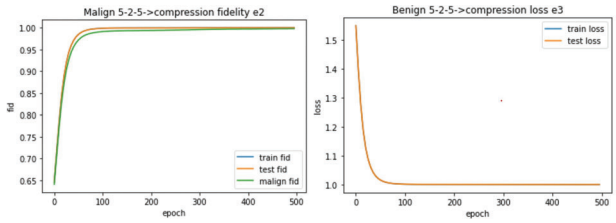

Fig. 8. Compression Fidelity & Loss of Second ansatz.

As shown in Fig. 6. a), the model initially compresses benign and anomalous data similarly. However, after a few epochs, a gap emerges in the encoding quality, revealing the anomalous cells.

As depicted in Fig. 7, histograms of encoding fidelity from simulations reveal a noticeable difference between the two categories, though some overlap persists. By setting a split value of 0.845, cells compressed with fidelity below this threshold are classified as benign, while well-compressed cells are deemed malign, achieving a **0.9** accuracy rate in identifying dangerous cells, i.e accuracy above **90%**. Fig. 7(a) and 7(b) compare the inference compression fidelity between Qiskit simulations and IBM QPU (hardware) results, respectively.

2) Second & Third ansatz: We extended our analysis by comparing the results of our framework, developed using the first ansatz, with those from prior works [23] and [24]. To ensure a fair comparison, we recreated their frameworks and followed their methodologies to obtain the corresponding results.

In the second ansatz, illustrated in Fig. 8, we adopted the circuit design proposed in [23], obtaining a split value of 0.99 and a malign classification accuracy of 0.24. For the third ansatz, also depicted in Fig. 9, we implemented the circuit design from [24], setting a split value of 0.96, a malign classification accuracy of 0.81. As evident from these results, our proposed technique demonstrates superior performance, providing higher detection accuracy and lower training and testing loss.

B. Credit Card Fraud Detection: Results and Insights

Training the e2 encoder on the credit card dataset achieved a compression fidelity of 0.89 ($\approx$ 90%) for non-fraudulent transactions and 0.66 for fraudulent transactions. For optimal performance in distinguishing between non-fraudulent and fraudulent transactions, it is desirable to maximize the compression fidelity of non-fraudulent transactions while minimizing the compression fidelity of fraudulent ones. These results demonstrate the encoder's capability to differentiate effectively between the two classes, aligning with the desired performance criteria (see Fig. 10).

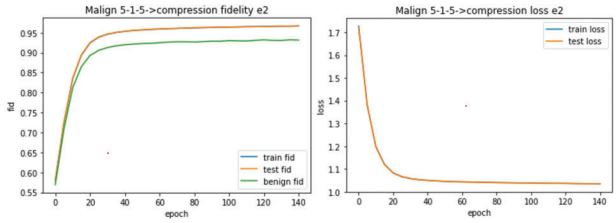

Fig. 9. Compression Fidelity & Loss of Third ansatz.

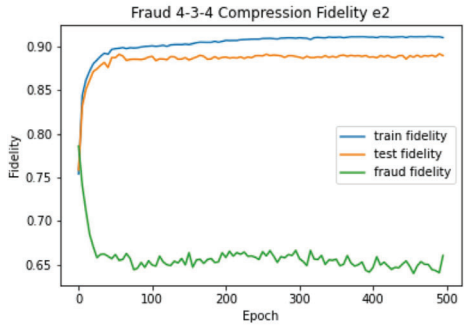

Fig. 10. Compression Fidelity of Financial Fraud Detection.

Fig. 11.(a) illustrates the compression fidelity results, highlighting the separation between fraudulent and non-fraudulent transactions. The histogram analysis underscores the importance of minimizing overlap between the distributions and maximizing the distance between their peaks to achieve optimal classification accuracy. Ideally, the non-fraudulent transactions should peak on the right side of the graph, while the fraudulent transactions peak on the left, ensuring effective distinction between the two categories.

Fig. 11.(b) presents the results obtained using IBM's 16-qubit Guadalupe quantum processor via Qiskit, employing the best parameters derived from training the e2 encoder. While the histogram structure aligns with the desired separation, the overlap between the distributions introduces challenges in accurately classifying transactions. This overlap results in a classification accuracy of 0.992 for non-fraudulent transactions, showcasing the model's limitations in this hardware implementation.

Fig. 11.(c) depicts the outcomes from Rigetti's Aspen-11 QPU, accessed through Amazon Braket, using the same optimal parameters derived from the e2 encoder. Here, the histogram separation is further reduced, limiting the model's capability to differentiate between fraudulent & non-fraudulent transactions. In comparison, the results obtained from the simulator in Fig. 11.(a) demonstrate superior histogram separation, establishing the simulator's outcomes as the most representative of the model's potential.

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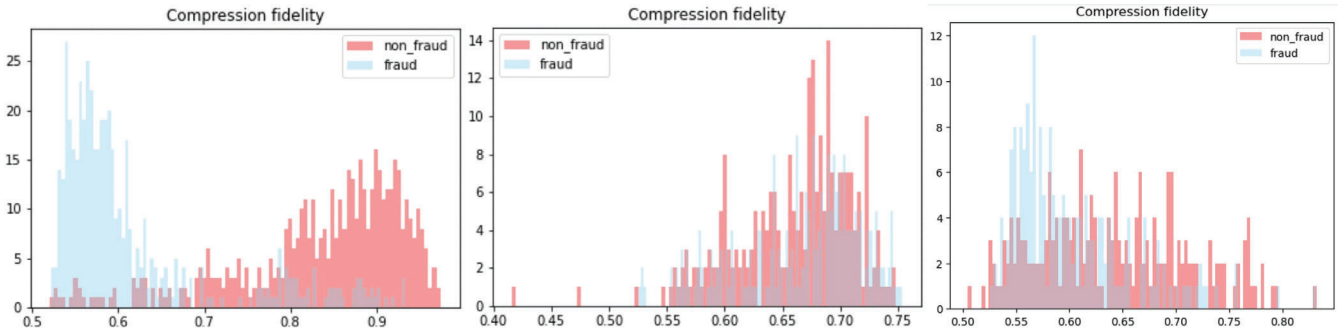

Fig. 11. Performance Evaluation: a) Simulator; b) IBM 16-Qubit Guadalupe (Qiskit Hardware); c) Rigetti Aspen-11 QPU (Amazon Braket Hardware).

V. Conclusion and Future Scope

In this work, we demonstrated the application of quantum autoencoders (QAEs) for anomaly detection through our model QEA-FAD, addressing two key challenges: distinguishing between malignant and benign cells in biological data, and identifying fraudulent transactions in financial datasets. Our simulation results show high accuracy in anomaly detection, with initial steps taken towards running our approaches on real hardware using IBM and Amazon platforms. We explored the potential of enhancing QAE performance through quantum entanglement, but found that improvements were more evident in interconnected encoder designs rather than entanglement alone. Future research will focus on optimizing the use of entanglement in QAE, identifying scenarios where it can enhance compression and recovery, and further advancing anomaly detection techniques. Moving towards real hardware and improving NISQ-era compatibility will be crucial to making our methods more robust and scalable for practical applications.

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